Zichen Zhao

173-2139-5871 | brucezhaozichen@126.com | BruceZhao0728 | brucezhao0728.github.io

EDUCATION

Shanghai Jiao Tong University (SJTU) Sep. 2023 – Jun. 2027 (Expected)
Bachelor of Engineering, School of Computer Science Shanghai, China

- **Coursework**: Design & Analysis of Algorithms, Compiler Principles, CTF Security Practice (Misc, Web & Pwn), AI Security, Data Structures, Mathematics Fundamentals of Information Security.
- **Honors**: 2024 SJTU "Premier Cybersecurity" Scholarship **(The sole sophomore recipient)**, 2024 & 2025 SJTU Excellence Scholarship, Second Prize in the 3rd Electrical & Electronic Engineering Innovation Competition.

RESEARCH EXPERIENCE

Lab of Cryptography and Computer Security, SJTU Apr. 2024 – Present
AI Software Protection System Based on Code Obfuscation Co-First Author

- Designed an automated code protection based on **TVM** and **LLVM IR** to shield localized AI models from reverse engineering, preventing unauthorized extraction of proprietary model architecture.
- Pioneered the theoretical expansion of Mixed Boolean-Arithmetic (MBA) from the integer domain to the floating-point domain, successfully obfuscating core AI operators (e.g., Conv) while maintaining **100% numerical consistency**.
- Engineered a custom LLVM Pass to implement multi-level obfuscation, allowing for tunable parameters such as recursion depth and MBA complexity to balance performance overhead and security strength.
- Quantified efficacy by developing a standardized metric system; experimental results showed a **>30% decrease in de-obfuscation success rates** and a **>40% decrease in reverse engineering success rates** compared to state-of-the-art works.
- Completed a research paper as **co-first author**, with related techniques under national invention patent application, and the manuscript prepared for submission to a CCF-A ranked international information security conference.

  Yikun Hu, Zichen Zhao, Peixiang Qin, Ziyi Zhou, Jiaping Gui, Yuandao Cai, and Wensheng Tang. *Protecting Floating-Point Computation for DNN Binaries with MBA Obfuscation*. 2026. arXiv: 2607.17603 [cs.CR]. URL: https://arxiv.org/abs/2607.17603

University Students' Innovative Program, SJTU Oct. 2023 – Oct. 2024
Design of a New Distributed Power Coordinated Controller Core Team Member

- Architected a decentralized **"Cloud-Edge" collaborative control framework** to address modeling complexities in modern power systems featuring multi-energy flow (Electricity/Gas/Heat) and massive distributed resources.
- Developed a distributed consensus algorithm based on Multi-Agent Systems and Graph Theory; derived the optimality condition for marginal cost/benefit alignment to ensure global convergence and power balancing.
- Optimized communication network reliability by implementing a genetic algorithm to evolve adjacency matrices, effectively enhancing the system's resilience against latency and structural failures.
- Implemented the distributed architecture on Raspberry Pi hardware using TCP/IP protocols for real-time data exchange between the cloud platform and microgrid controllers.
- Awarded **second prize** in the 3rd Electrical & Electronic Engineering Innovation Competition.

Cyberspace Security Workstation, SJTU Jun. 2021 – Oct. 2021
Research on Home-Used Wi-Fi Network Security Research Individual

- Conducted cyberattacks on home-used Wi-Fi systems using **Kali Linux** and demonstrated the critical security flaws.
- Executed advanced penetration techniques, including ARP packet injection for WEP cracking and Dictionary Attacks on WPA2 four-way handshakes to expose PSK vulnerabilities.
- Simulated complex attack vectors such as "Evil Twin" Access Points and De-authentication floods combined with MAC address spoofing to hijack sessions and bypass MAC filtering.
- Selected for the **Excellent Thesis Defense** at the SJTU Cyberspace Security Workstation

COURSE PROJECTS

Database Principles & Security, SJTU Sep. 2025 – Oct. 2025
AI-Powered Vulnerability Analysis System Based on Vector Databases Team Leader

- Architected a hybrid-engine vulnerability detection system utilizing a **dual-database architecture**: Chroma (Vector DB) for semantic similarity searches of code slices and MySQL (Relational DB) for structured management of CVE/CWE datasets.
- Implemented advanced **program slicing and static analysis** using Tree-Sitter to parse ASTs and extract data-flow/control-flow dependencies across multiple programming languages.
- Developed a semantic-structural embedding engine utilizing **CodeT5+** and **GraphCodeBERT** to capture dual-feature vectors of program slices, enabling high-precision similarity matching against known CVE/CWE patterns.
- Engineered a natural language interface that automates SQL query generation via cloud-based LLM APIs and integrated a responsive Django/Bootstrap 5 web platform for real-time security auditing.

InfoSec SciTech Innovation, SJTU Jun. 2025 – Jul. 2025
Linux Kernel-Module-Based Packet Filtering Firewall Team Leader

- Designed and implemented a high-performance, kernel-level firewall leveraging the Netfilter framework to intercept and analyze network traffic at key hooks while eliminating frequent context switches.
- Developed a custom **Linux Kernel Module** utilizing zero-copy packet interception and deep packet inspection to enforce stateful filtering and protocol-based access control.
- Architected a robust cross-layer communication system using **Netlink** sockets and character devices to synchronize real-time security events and dynamic rule updates between the kernel and a dedicated user-space daemon.
- Built a comprehensive management suite including a CLI tool with iptables-like syntax and a GUI dashboard for real-time traffic monitoring and anomaly visualization.
- Awarded the highest grade among all course projects for exceptional system stability and architectural complexity.

SKILLS & HOBBIES

Languages : Python, C/C++, LLVM
Tools : Docker, IDA Pro, Raspberry Pi
Hobbies : Piano (Level 10), Music Theory (Level A), Badminton (Top 16, Men’s Singles, SJTU Freshman Cup)